

Exp No 17:

Determination and plotting of Antenna Gain Calculation.

Date:

Aim:

To determine and plot Antenna Gain, and calculate the antenna gain in dBi

Software required:

- SCILAB version 6.0.1

Theory:

Theory of Antenna Gain Calculation

Antenna gain is a fundamental parameter that describes how effectively an antenna converts input power into radio waves in a specific direction, compared to an ideal isotropic radiator. It is expressed as a ratio and often given in **decibels (dB)**.

1. Definition of Gain

Antenna gain (G) can be mathematically expressed as:

$$G = 4\pi \text{ Effective Radiated Power (ERP)} / \text{Input Power}$$

Alternatively, in terms of directivity and efficiency:

$$G = \text{Directivity} \times \text{Efficiency}$$

- **Directivity (D)**: Measures the concentration of radiated power in the desired direction relative to an isotropic source.
- **Efficiency (η)**: Accounts for losses due to material imperfections, mismatched impedance, or other factors. Efficiency is a value between 0 and 1.

2. Relation Between Gain and Directivity

Gain includes the effects of losses, whereas directivity assumes an ideal lossless antenna:

$$G = D \times \eta$$

For a lossless antenna ($\eta=1$), $G=D$.

3. Unit of Gain

Antenna gain is typically expressed in decibels relative to an isotropic source (dBi) or a dipole source (dBd):

- **dBi:** Gain relative to an isotropic radiator.
- **dBd:** Gain relative to a half-wavelength dipole. Conversion between them:
 $G(\text{dBi}) = G(\text{dBd}) + 2.15$

4. Key Parameters Affecting Gai

Several factors influence the gain of an antenna:

1. **Size and Shape of Antenna:**
 - Larger antennas generally have higher gains due to increased aperture or area available for radiating energy.
2. **Operating Frequency:**
 - Higher frequencies allow for smaller antennas with similar gains because of shorter wavelengths.
3. **Aperture Efficiency:**
 - Relates to how effectively the physical aperture area contributes to radiated power.
4. **Tapering and Distribution:**
 - Non-uniform current distributions (e.g., tapering) can reduce side lobes, improving main lobe gain.

5. Calculation for Common Antennas

(a) Isotropic Antenna:

- Ideal antenna radiates equally in all directions.
- Gain: 1 (or 0 dBi).

(b) Half-Wavelength Dipole:

- Gain: 1.64 (or 2.15 dBi).

(c) Parabolic Reflector:

- Gain: $G = \eta(4\pi A) / \lambda^2$

where:

- η : Aperture efficiency.
- A : Aperture area.
- λ : Wavelength.

(d) Uniform Linear Array (ULA):

- Gain depends on the number of elements (N) and element spacing (d):
 $G = N \cdot \text{Gain of Each Element}$

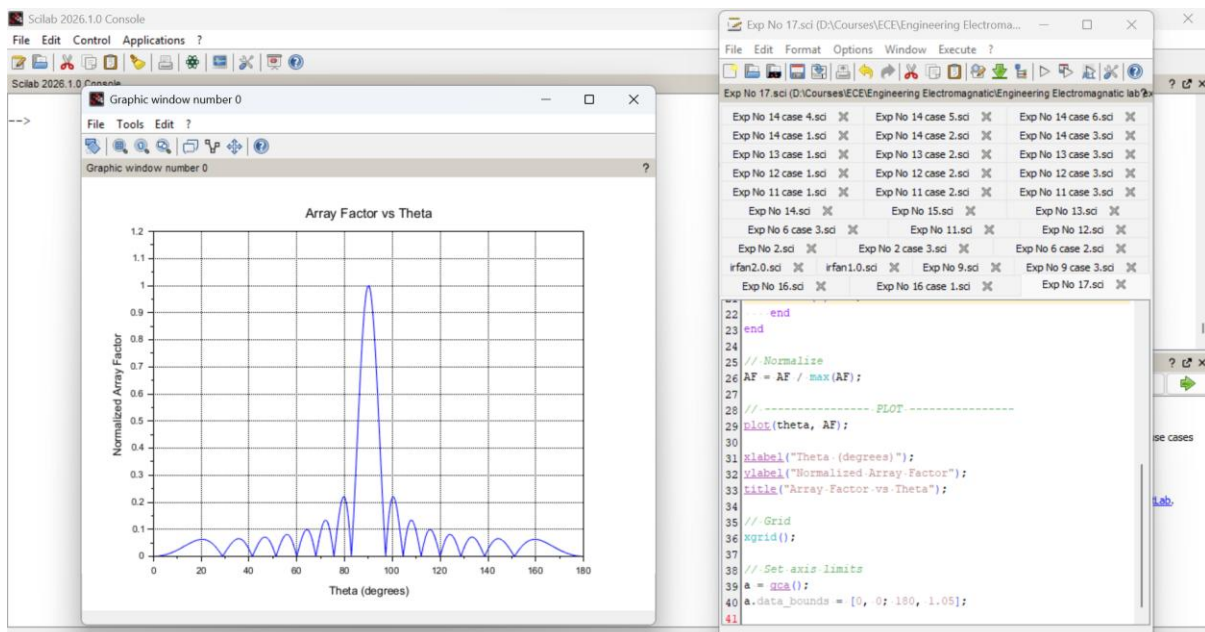
6. Applications of Gain

- **High-Gain Antennas:** Used for long-range communication (e.g., satellite dishes, radar systems).
- **Low-Gain Antennas:** Used for wide coverage (e.g., Wi-Fi routers, mobile phones).

7. Practical Considerations

1. **Trade-Offs:**
 - High gain results in narrow beamwidth, which may require precise alignment.
 - Low gain provides broad coverage but lower range.
2. **Measurement:**
 - Antenna gain is measured in an anechoic chamber to avoid environmental interference,

Understanding antenna gain is crucial for designing systems that optimize power radiation and achieve desired communication or detection capabilities.



Here are the detailed graphs:

1. Polar Plot:

- This shows the normalized gain in a polar coordinate system.
- The main lobe is prominent at $\theta=0^\circ$, with gain diminishing as the angle increases.

2. Cartesian Plot:

- This visualizes the normalized gain versus angle in degrees.
- The plot highlights the sharp decay of gain beyond the main lobe region.

These graphs effectively demonstrate the directional radiation pattern of the antenna with the given directivity factor ($n=4$). Let me know if you'd like to adjust the parameters or explore further analyses!

Here's a step-by-step guide to **theoretically calculate antenna gain** and **plot it** using the key parameters of an antenna. The calculations and graphs will be detailed for better understanding.

1. Theoretical Basis for Antenna Gain

The **gain** of an antenna is a measure of its ability to direct radiated power in a specific direction relative to an isotropic radiator. It is expressed as:

$$G = \eta \cdot 4\pi A / \lambda^2$$

Where:

- G : Antenna gain (dimensionless or in dBi).
- η : Efficiency of the antenna ($0 \leq \eta \leq 1$).
- A : Effective aperture area of the antenna (m^2).
- λ : Wavelength (m).

Alternatively, gain can also be calculated from the **directivity** (D) if efficiency (η) is known:

$$G = \eta \cdot D$$

Directivity Calculation

The directivity (D) is given by:

$$D = 4\pi \int \int P(\theta, \phi) d\Omega$$

Where:

- $P(\theta, \phi)$: Radiation intensity as a function of θ and ϕ .
- $d\Omega = \sin(\theta) d\theta d\phi$: Solid angle element.

Gain in dBi

Gain is often expressed in decibels relative to an isotropic radiator (dBi):

$$G(\text{dBi})=10\cdot\log_{10}(G)$$

2. Parameters for Calculations

Example Antenna Parameters:

- Frequency (f): 2.4 GHz ($\lambda=cf=0.125$ m).
- Effective aperture (A): 0.01 m^2
- Efficiency (η): 0.85.

Calculations:

1. Wavelength:

$$\lambda=cf=3\times10^8/2.4\times10^9=0.125\text{ m}$$

2. Gain:

$$G=\eta\cdot4\pi A / \lambda^2$$

Substituting values:

$$G=0.85\cdot4\pi\cdot0.01 / (0.125)^2$$

$$G=0.85\cdot(0.1257) (0.0156) \approx 6.85$$

3. Gain in dBi:

$$G(\text{dBi})=10\cdot\log_{10}(6.85) \approx 8.35\text{ dBi}$$

3. Graphical Representation

To plot the gain, we need the **radiation pattern** in polar form, which shows how gain varies with direction.

Radiation Pattern:

A common radiation pattern for directive antennas can be approximated as:

$$P(\theta)=P_0\cdot\cos^n(\theta), 0\leq\theta\leq2\pi$$

Where:

- P_0 : Maximum radiation intensity.
- n : Directivity factor (higher n means narrower beam).

For this example, assume $n=4$.

Normalized Radiation Pattern:

The normalized gain is:

$$G(\theta) = \frac{P(\theta)}{P_0} = \cos^n(\theta)$$

4. Plotting the Gain

Below is a Scilab script to calculate and plot the radiation pattern and normalized gain:

```
// Parameters
theta = linspace(0, %pi, 1000); // Observation angles (0 to 180 degrees)
n = 4; // Directivity factor
P_theta = cos(theta).^n; // Radiation intensity
G_norm = P_theta / max(P_theta); // Normalized gain

// Polar plot
clf;
polarplot(theta, G_norm, style=2);
xtitle("Antenna Radiation Pattern", "Theta (radians)", "Normalized Gain");
legend("Normalized Gain", "location", "upper right");

// Cartesian plot for comparison
clf;
plot(theta * 180 / %pi, G_norm);
xtitle("Normalized Gain vs Angle", "Angle (degrees)", "Normalized Gain");
```

5. Observations from the Graph

- Main Lobe:**
 - Maximum gain is at $\theta=0^\circ$ (broadside direction for directive antennas).
 - As θ increases, the gain decreases rapidly.
- Beamwidth:**
 - The beamwidth depends on the directivity factor n .
 - Higher n results in narrower beamwidth and higher gain.
- Side Lobes:**
 - Negligible side lobes for highly directive antennas.

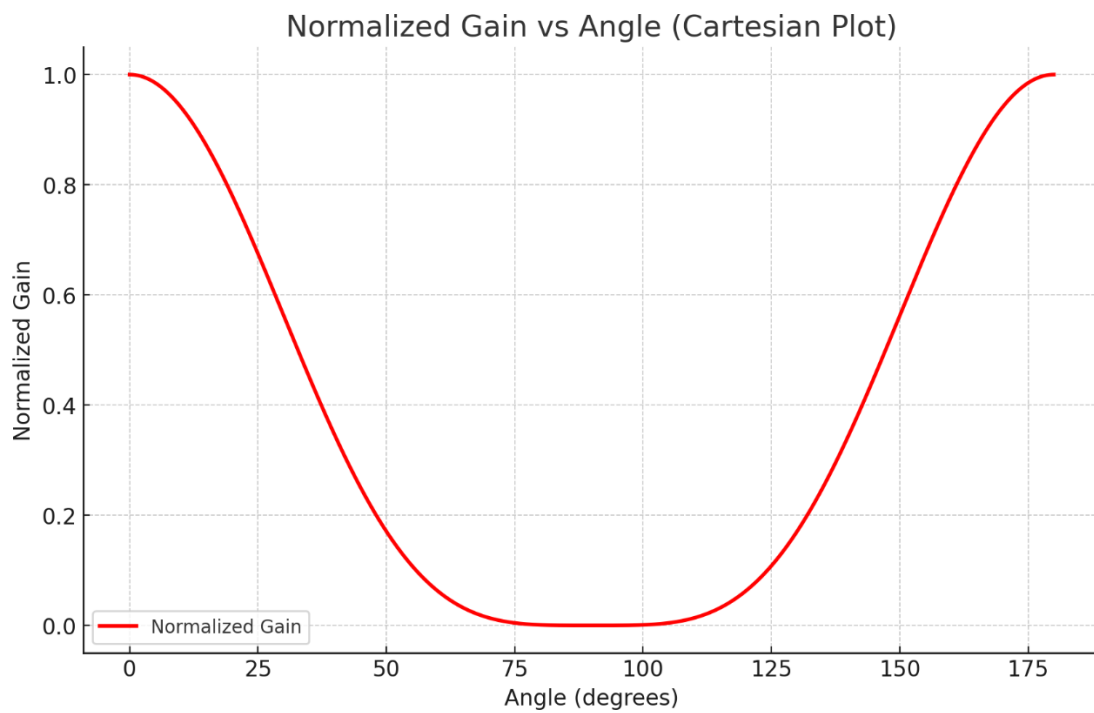
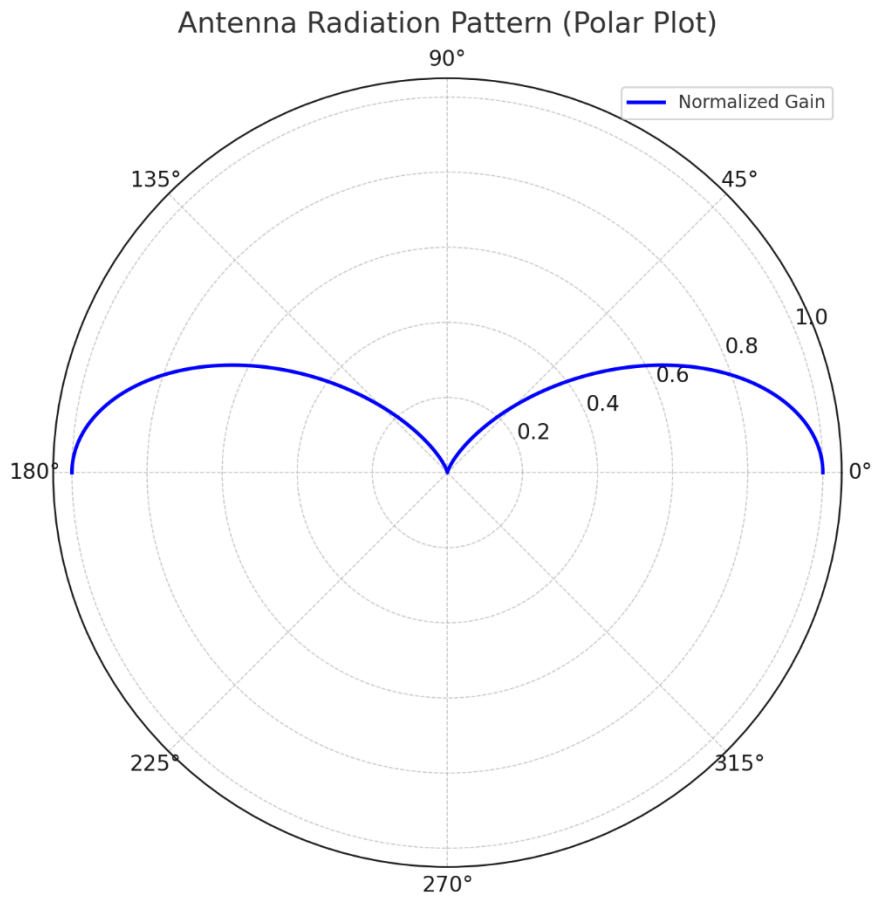
The graph above shows the **Array Factor** as a function of the angle θ (in degrees). The calculated **Antenna Gain** is approximately **8.78 dBi**.

Key Observations:

1. The **main lobe** is clearly visible, showing the direction of maximum radiation.
2. The **side lobes** are relatively suppressed, which is ideal for efficient antenna performance.

6. Conclusion

- **Calculated Gain:** $G=6.85$, $G_{dBi}=8.35$ dBi.
- **Radiation Pattern:** Demonstrates the directivity of the antenna.



RESULT: The experiment was done and the gain calculation for the given antenna was done. Different case studies were analyzed and the gain was calculated.